



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
PRE-SCORE
22-10-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

Time : 3 Hours

PAPER-1 (OPTIONAL)

Maximum Marks : 186

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

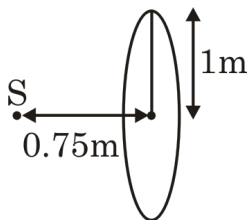
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

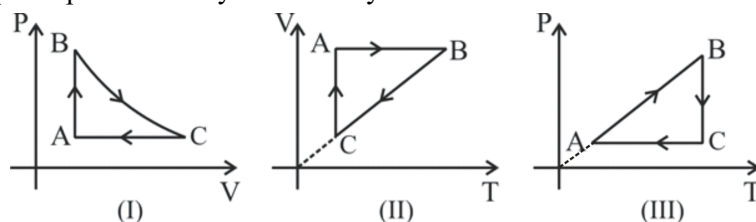
1. A black thin disc of radius 1m is kept 0.75 m away from a point source of power $56.7 \pi \text{ W}$. If the disc is highly conducting, what is its temperature in steady state? Surrounding is at 0 K.



- (A) 100 K
 (B) 141 K
 (C) 70.5 K
 (D) 282 K
2. A non-linear triatomic gas is filled inside a vessel. If ' α ' fraction of moles dissociate into individual atoms, then average degree of freedom for the mixture is : (neglect vibrational degrees of freedom)

- (A) $\frac{3\alpha + 6}{\alpha + 1}$
 (B) $\frac{\alpha + 6}{2\alpha + 1}$
 (C) $\frac{3\alpha + 6}{\alpha + 2}$
 (D) $\frac{3\alpha + 6}{2\alpha + 1}$

3. An ideal gas goes through a cycle. Starting from state A, it is first heated at constant volume up to state B, then it expands to state C without a change in temperature and is finally isobarically compressed again to initial state A. Use p , V and T to denote the pressure, volume and temperature of the gas. Which of the following graphs represent the cycle correctly?



- (A) Graph I and II only.
 (B) Graph I and III only.
 (C) Graph II and III only.
 (D) All three graphs.
4. On a cold winter's day, there are three identical, non-insulated wooden boxes in front of the house, each filled with the same amount of ice at a temperature of 0.0°C . To melt the ice, an electric heating element is placed in each of the boxes. The heating elements are identical, but are operated with different voltages. In the first box, the heating element is operated with a voltage of 80 V . All of the ice in the box will then melt in 20.0 minutes. A voltage of 120 V is applied to the heating element of the second box, whereupon the ice completely melts in just 4.0 minutes. In the third box, a voltage of 40 V is used for the heating element. The heating elements are designed in such a way that they heat the entire mass of ice in the respective box at the same time. Assume that the meltwater is not being heated by the heating element. Which of the following statements is then true for the melting of the ice in the third box?
- (A) It takes about 80 minutes to melt all of the ice in the third box.
 (B) It takes about 100 minutes to melt all of the ice in the third box.
 (C) It will take about 130 minutes to melt all of the ice in the third box.
 (D) With the voltage used, it is not possible to melt all of the ice.

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

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5. A process for monoatomic gas is defined by $V = aT^2$, where a is a constant. In this process temperature of 1 mole of a gas is increased by 100 K; then (V = Volume, T = Temperature)

- (A) Molar heat capacity of this process is negative.
(B) The work done by the gas in the process is 250 R.
(C) The work done by the gas in the process is 200 R.
(D) Heat given to the gas is 350 R.

6. Two spherical black bodies A and B, having radii r_A and r_B where $r_B = 2r_A$ emit radiation with peak intensities at wavelength 400 nm and 800 nm respectively. If their temperature are T_A and T_B respectively in Kelvin scale, their emissive powers are E_A and E_B , and energy emitted per second are P_A and P_B then

(A) $\frac{T_A}{T_B} = 2$

(B) $\frac{P_A}{P_B} = 4$

(C) $\frac{E_A}{E_B} = 8$

(D) $\frac{E_A}{E_B} = 4$

7. A piece of ice having temperature 0°C is thrown into a thermally insulated container, filled with water of volume 5 dm^3 at a temperature of 50°C . Match the final amount of water in the container if after the ice melts the temperature of the system is $T^{\circ}\text{C}$?

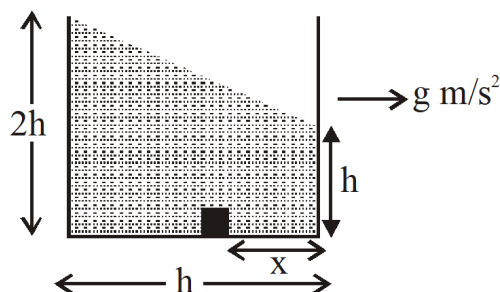
(A) $T = 15, V = \frac{35}{19} \text{ dm}^3$

(B) $T = 0, V = \frac{65}{8} \text{ dm}^3$

(C) $T = 5, V = \frac{130}{17} \text{ dm}^3$

(D) $T = 10, V = \frac{100}{9} \text{ dm}^3$

8. A vessel is completely filled by a liquid of density ρ as shown in figure having acceleration g in horizontal direction. After getting equilibrium condition a particle of density $\frac{\rho}{2}$ is released from bottom of vessel at distance 'x' as shown in figure such that it reaches to water-air interface after time $\sqrt{\frac{2h}{g}}$. Assume all collisions with wall are completely elastic and walls are smooth. Which of the positions satisfy the given situation :-



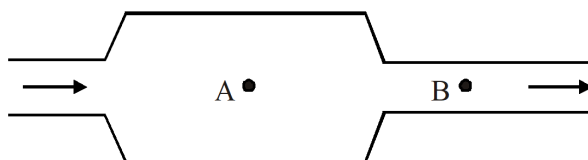
(A) $x = 0$

(B) $x = h$

(C) $x = \frac{h}{9}$

(D) $x = \frac{h}{4}$

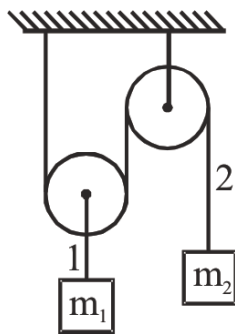
9. The diagram below shows a cross-sectional view of a cylindrical pipe of varying diameter.



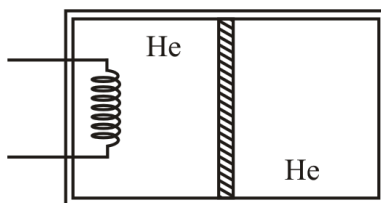
If an ideal fluid is flowing through the pipe, find the true statement(s).

- (A) Volume flow rate is same at A & B
 (B) The pressure is lower at point B than at point A.
 (C) The volume flow rate is greater at point A than at point B.
 (D) The flow speed is greater at point B than at point A.

10. In the figure shown the system is in equilibrium. In this system we have used two light wires 1 and 2. The Young's modulus and radius of the wire 1 are two times and $2^{1/4}$ times respectively that of wire 2, then.



- (A) Elastic energy density in wire 1 is equal to that of in wire 2
 (B) Elastic energy density in wire 1 is greater than that of in wire 2
 (C) Elastic energy density in wire 1 is less than that of in wire 2
 (D) Stress in wire 1 is greater than stress in wire 2.
11. An ideal heat engine has two sources of finite thermal capacity $C = 500 \text{ J/K}$ each, with initial temperature $T_H = 625 \text{ K}$ and $T_C = 289 \text{ K}$. The power of the engine decreases exponentially in time according to the law $P = P_0 e^{-k\tau}$, where $P_0 = 960 \text{ W}$ is the initial power, and τ is the time elapsed.
- (A) The temperatures of the two sources, when the engine ceases to function is 457 K .
 (B) The total work done by the engine is 32 kJ .
 (C) $k = 0.6 \text{ K/sec}$.
 (D) Initially, the temperature T_H decreases at rate of $\frac{25}{7} \text{ K/sec}$.
12. A Thermally insulated cylindrical vessel is divided into two equal parts by a non-conducting piston that can move without friction. At the initial moment in the left and right parts of the vessel contains one mole of helium at the same temperature. Heat was supplied to the left side of the vessel using a heater. At the same time, the temperature of the helium in it increased by a small amount ΔT . The temperature changes by ΔT_2 on the right side of the vessel and the amount of heat Q is transferred by the heater to the system.



(A) $\Delta T_2 = \frac{\Delta T}{4}$

(B) $\Delta T_2 = \frac{\Delta T}{2}$

(C) $Q = \frac{9R\Delta T}{4}$

(D) $Q = \frac{15R\Delta T}{8}$

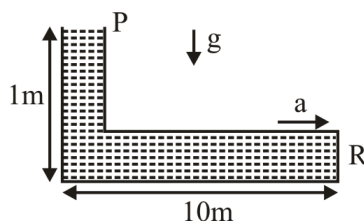
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

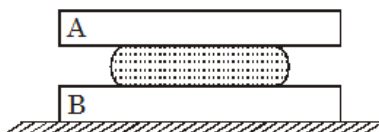
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

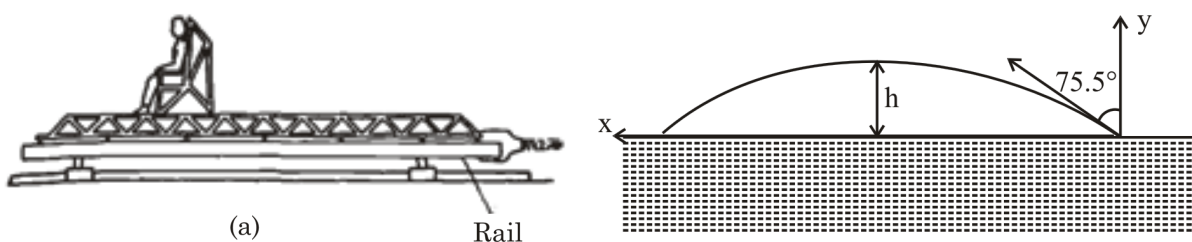
1. A glass bulb with volumetric expansion coefficient $\alpha = 1.50 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$ is weighed in liquid at temperatures $t = 10^\circ\text{C}$ and $t_1 = 20^\circ\text{C}$. The weights of the displaced liquids are $p = 100.01 \text{ g}$ and $p_1 = 100 \text{ g}$ respectively. Find the volumetric expansion coefficient α_1 (in the units of $10^{-5} \text{ } ^\circ\text{C}^{-1}$) of the liquid in the temperature interval from t to t_1 .
2. A uniform tube PQR is in the form of right angle as shown in the figure. It is closed at R and has a small cross-sectional area A. It is filled with water till point P and is accelerating horizontally with acceleration "a" as shown in the figure. Find the acceleration a (in m/s^2) so that pressure at R is atmospheric. (Take $g = 10 \text{ m/s}^2$)



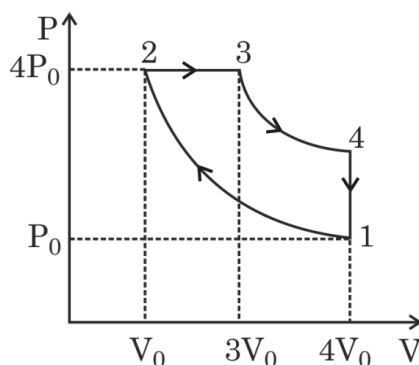
3. A drop of mercury is present between two glass plates each of mass 50 gm. Separation between the plates is negligible as compared to spread of the mercury between the plates. What should be the mass of a block (in kg) which should be placed over the plate A so that the separation between the plates becomes half of the initial separation. Assume contact angle between mercury and glass is 180° .



4. A rocket-powered sled is used in the training of astronauts as shown in figure. For braking, scoops are lowered to deflect water from a stationary tank of water. To what height h (in m) does a sled traveling at 30 m/s deflect water? Take $\sin 14.5^\circ = 0.25$



5. There's a one mole diatomic ideal gas following a thermodynamic cycle. The process consists of an isothermal compression followed by an isobaric heating, which is followed by a polytropic process and isochoric. The molar capacity of gas during polytropic process is $C = 0.5R$. Find the efficiency of this cycle in percentage. Take $\ln 2 = 0.7$



6. A coin is frozen into a piece of ice weighing 130 g and density 900 kg/m^3 . The coin weighs 10 g and density 9000 kg/m^3 . This piece of ice with the coin has temperature 0°C , placed in a vessel containing 400 ml of water with some initial temperature t . Heat exchange with the surrounding can be neglected. The ice and coin initially float without touching the bottom of the vessel. Specific heat capacity of water $4200 \text{ J/(kg} \cdot ^\circ\text{C)}$, specific heat of melting of ice is 330 kJ/kg , water density 1000 kg/m^3 , acceleration free fall 10 m/s^2 . What should be the minimum initial water temperature so that remaining piece of ice along with a coin sank to the bottom after it attains thermal equilibrium? Express your answer in degrees celsius.

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

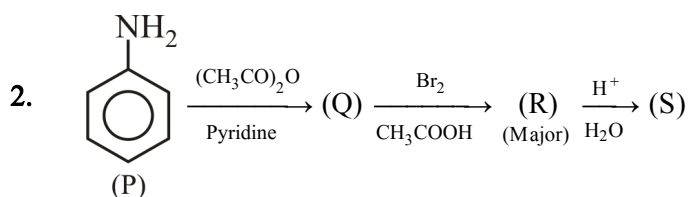
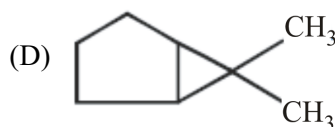
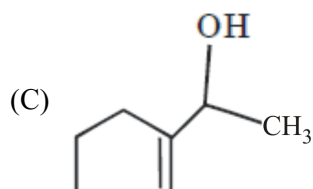
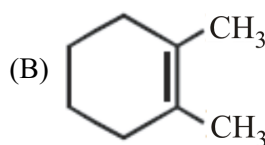
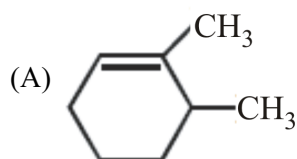
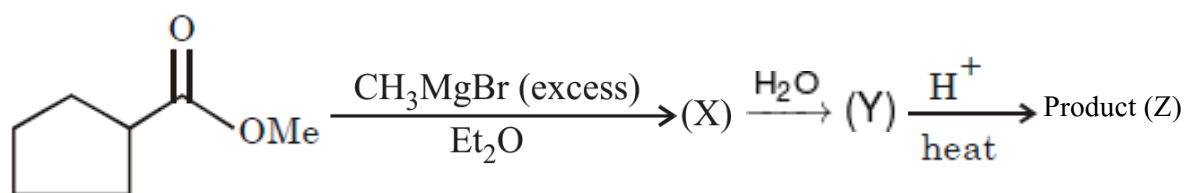
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Negative Marks : -1 In all other cases

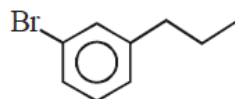
1. Major product "Z" obtained in given reaction is ?



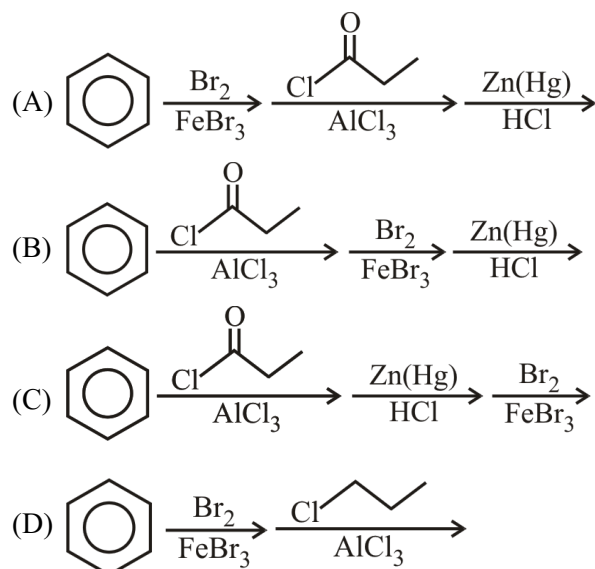
Choose the incorrect option ?

- (A) (S) is 4-bromoaniline
(B) (Q) is Acetanilide
(C) Basic strength order is : (Q) > (P) > (R)
(D) Basic strength order is : (P) > (R) > (Q)

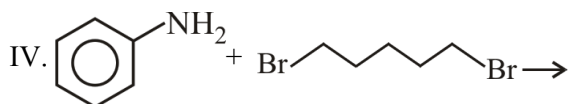
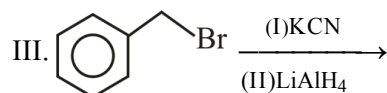
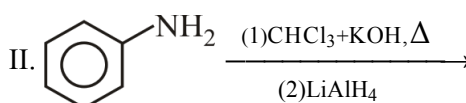
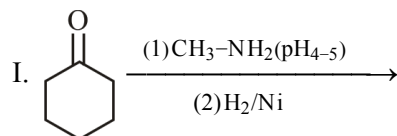
3. Which of the following reaction sequence would be the best for synthesizing the compound, 1-bromo-3-propylbenzene ?



1-Bromo-3-propylbenzene



4. Product of which of the following reaction gives carbyl amine test.



- (A) I and II
(B) I, II, III
(C) I and III
(D) III only

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

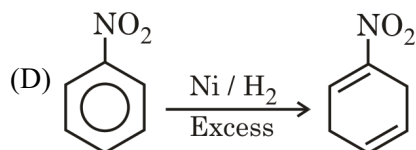
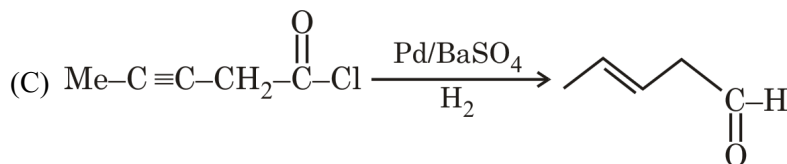
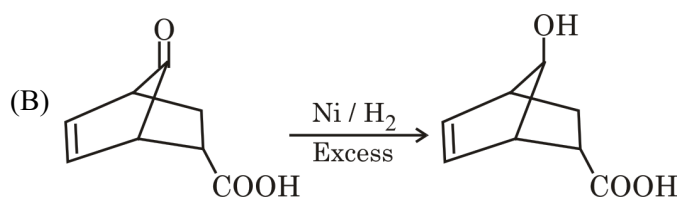
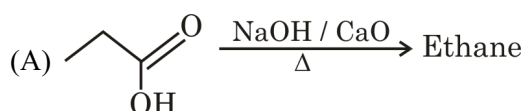
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

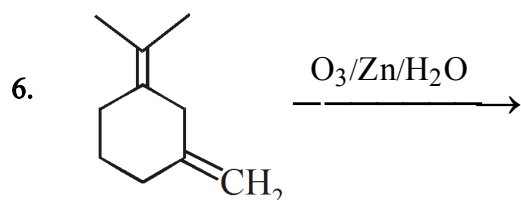
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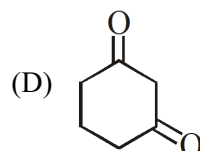
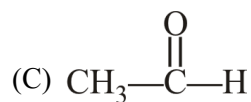
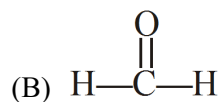
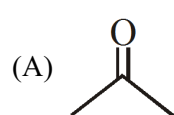
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5. Identify reactions incorrectly matched with their final products?

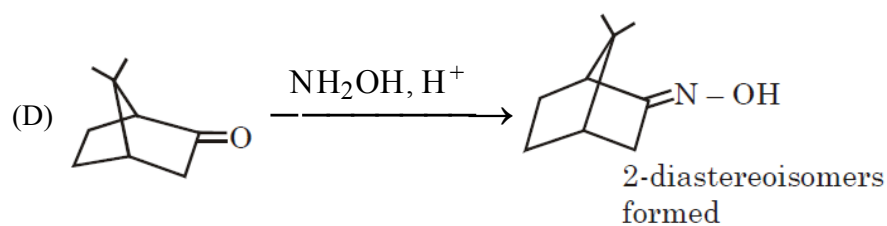
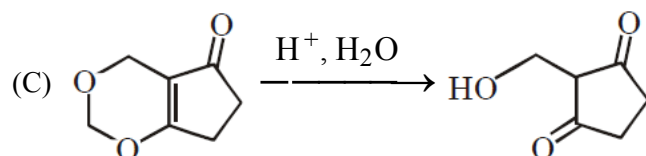
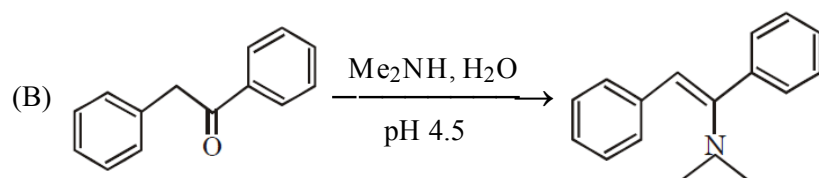
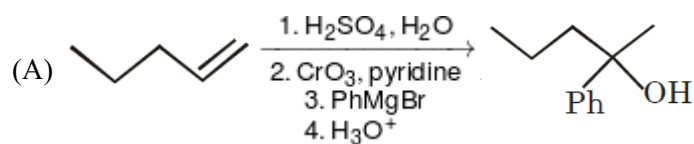




Products which is / are formed in above reaction

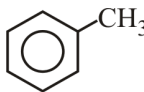


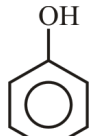
7. Identify reactions correctly matched with their major products :

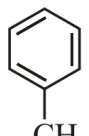


8. Which are Correct statement(s) in the following?

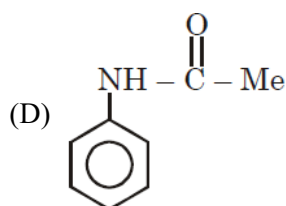
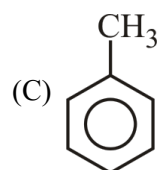
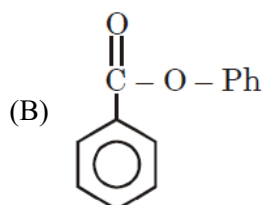
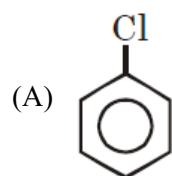
(A) Rate of sulphonation of C_6H_6 and C_6D_6 are equal.

(B) Rate of halogenation of  is faster than .

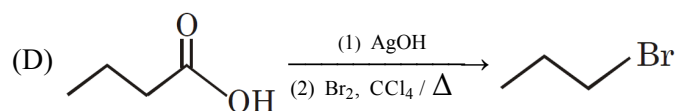
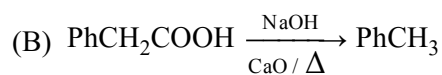
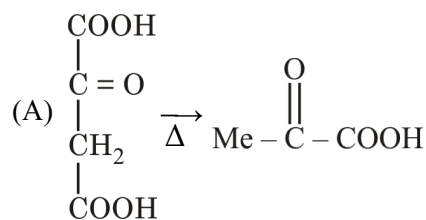
(C)  on reaction with Br_2 / H_2O forms tribromoderivative.

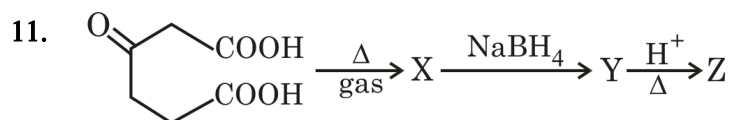
(D)  on reaction with Cl_2 at $500^\circ C$ will give electrophilic aromatic substitution reaction.

9. Which of the following will react at rate faster than benzene during aromatic electrophilic substitution?



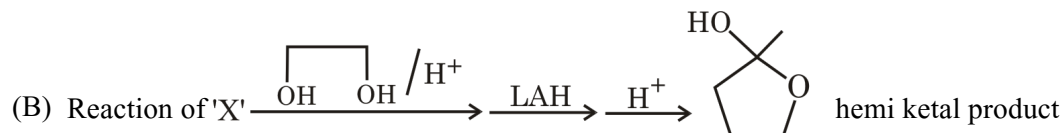
10. Select reaction(s) with correct major product :





Identify correct statement for above reaction sequence ?

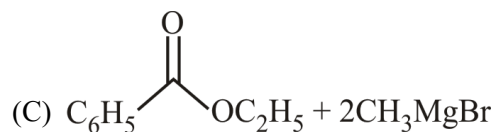
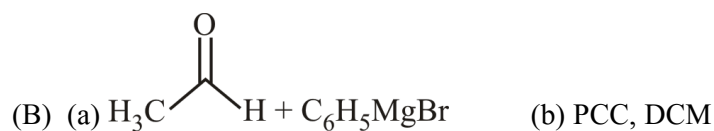
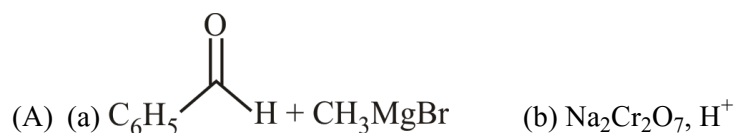
(A) 'Z' product is sweet smelling substance



(C) 'Y' product liberate CO_2 gas with aqueous NaHCO_3 and also soluble in aqueous NaOH

(D) 'Z' product is resolvable

12. Which of the following conditions or reaction sequence will NOT give acetophenone as the major product ?



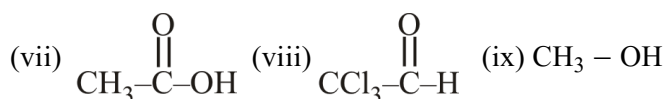
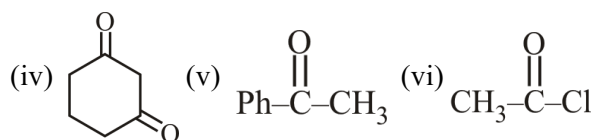
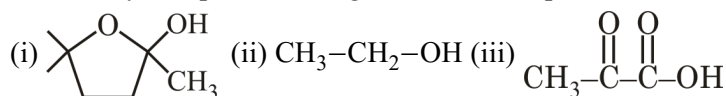
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

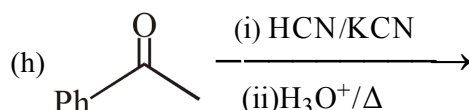
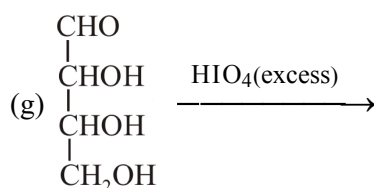
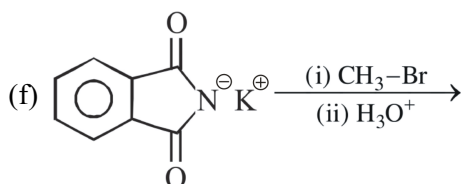
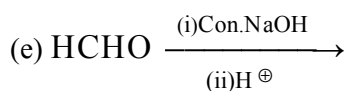
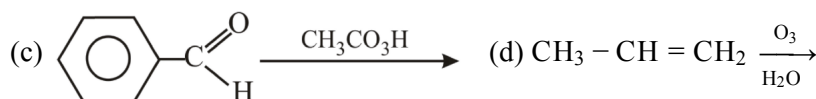
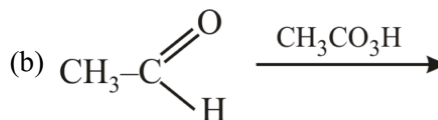
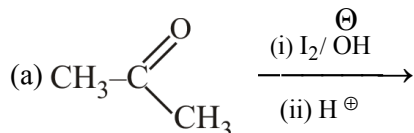
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

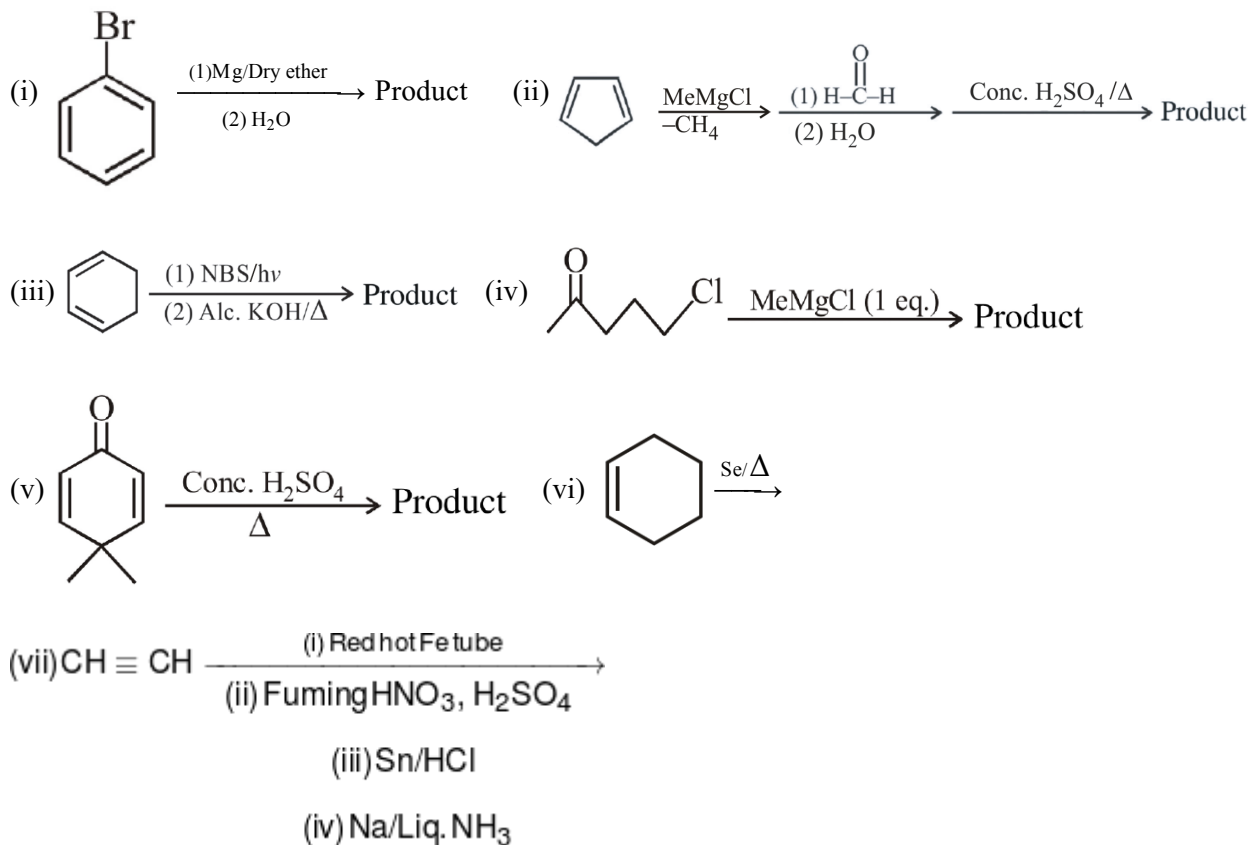
1. How many compounds will give Anesthesia product with $\text{Cl}_2 / \text{NaOH}$:



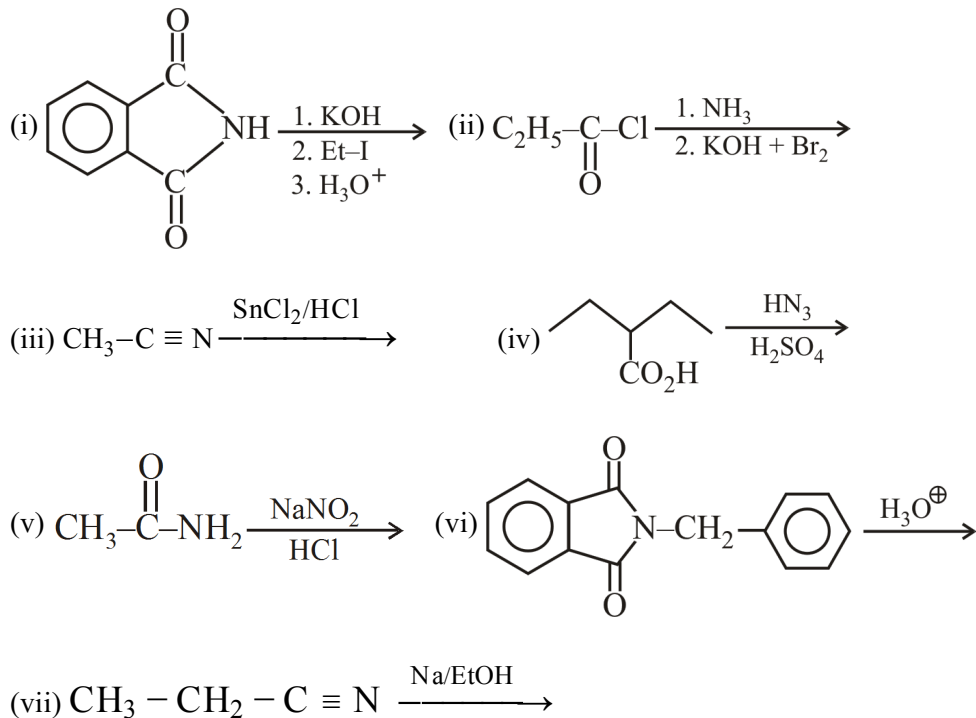
2. Find total number of reaction in which acetic acid is one of product.

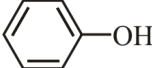


3. Total number of reactions which give aromatic product as a major final product?

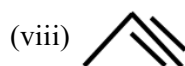
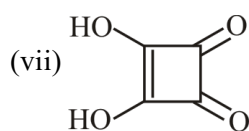
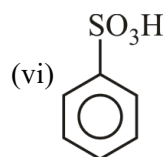
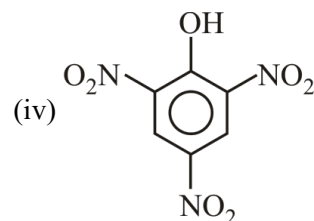
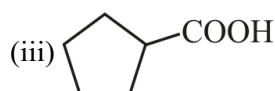
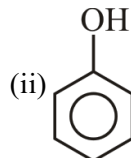
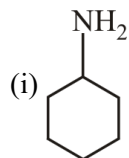


4. How many reactions will give 1° amine as final product :



5. How many of the following can be used to differentiate between $\text{CH}_3\text{—CH}_2\text{—OH}$ and -OH
- (a) NaOI (b) Neutral FeCl_3 (c) Azo-dye test (d) Libermann nitroso test
(e) Na (f) NaHCO_3

6. Find number of compound which are soluble in NaOH as well as NaHCO_3 ?



PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. $\cos \frac{\pi}{7} - \cos \frac{2\pi}{7} + \cos \frac{3\pi}{7}$ is
- (A) $\frac{1}{2}$ (B) $-\frac{3}{2}$
(C) $-\frac{1}{4}$ (D) $\frac{1}{8}$
2. Number of solution of $\frac{\tan \theta}{\sqrt{3}} + \frac{\tan 2\theta}{\sqrt{3}} + \tan \theta \tan 2\theta = 1$ in $\theta \in (0, 2\pi)$
- (A) 0 (B) 1
(C) 4 (D) 6
3. If lines L_1, L_2 are the member of family $(x - 2) + \lambda(y + 1) = 0$ and $(x - 5) + \mu(y - 3) = 0$ respectively (where λ and μ are parameter). Then maximum possible perpendicular distance between L_1 and L_2 is
- (A) $\frac{2}{3}$ (B) 3
(C) $\frac{7}{2}$ (D) 5
4. P is a point on the line $y + 2x = 1$ and Q and R are two points on the line $3y + 6x = 6$ such that the triangle PQR is an equilateral triangle. Area of this triangle is equal to
- (A) $\frac{1}{15}$ (B) $\frac{1}{5\sqrt{3}}$
(C) $\frac{1}{3\sqrt{5}}$ (D) $\frac{1}{2\sqrt{15}}$

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. If $\sin x \cos^5 x > \cos x \sin^5 x$ then x lies in

(A) $\left(0, \frac{\pi}{8}\right)$

(B) $\left(\frac{\pi}{8}, \frac{\pi}{4}\right)$

(C) $\left(\frac{\pi}{2}, \frac{5\pi}{8}\right)$

(D) $\left(\frac{5\pi}{8}, \frac{3\pi}{4}\right)$

6. If which of the following is/are correct

(A) If $\cos \theta = \frac{\left(a + \frac{1}{a}\right)}{2}$ then $\cos(3\theta) = \frac{\left(a^3 + \frac{1}{a^3}\right)}{2}$

(B) If $\cos x + \cos y + \cos z = 0$ then $\cos 3x + \cos 3y + \cos 3z = 12 \cos x \cos y \cos z$

(C) $\frac{(\cos 9^\circ + \sin 9^\circ)}{(\cos 9^\circ - \sin 9^\circ)} = \tan(54^\circ)$

(D) If $\sin \theta$ is geometric mean of $\sin \phi$ and $\cos \phi$ then $\cos 2\theta = 2\cos^2\left(\frac{\pi}{4} + \phi\right)$

7. Let $y = \frac{1}{\left(\sin^4 x + \cos^4 x + \frac{\sin 2x}{2} + 1\right)}$ then y can be
- (A) $\frac{1}{4}$ (B) $\frac{8}{17}$
(C) 1 (D) 0.8
8. All values of $x \in \left(0, \frac{\pi}{2}\right)$ such that $\frac{\sqrt{3}-1}{\sin x} + \frac{\sqrt{3}+1}{\cos x} = 4\sqrt{2}$ are
- (A) $\frac{\pi}{15}$ (B) $\frac{\pi}{12}$
(C) $\frac{11\pi}{36}$ (D) $\frac{3\pi}{10}$
9. If $a \cos x + \cot x + 1 = \operatorname{cosec} x$ then the possible values of x can be
- (A) $x = n\pi, a \in \mathbb{R}, n \in \mathbb{I}$
(B) $x = (4n+1)\frac{\pi}{2}, a \in \mathbb{R}, n \in \mathbb{I}$
(C) $x = \frac{1}{2}\sin^{-1}\left(\frac{4(a+1)}{a^2}\right), a \in (-\infty, 2-2\sqrt{2}] \cup [2+2\sqrt{2}, \infty) - \{-1\}$
(D) none of these
10. The equation of the circle which touches the axes of coordinates and the line $\frac{x}{3} + \frac{y}{4} = 1$ and whose centre lies in the first quadrant is $x^2 + y^2 - 2cx - 2cy + c^2 = 0$ where c is
- (A) 1 (B) 2
(C) 3 (D) 6
11. If $\tan \theta_1, \tan \theta_2, \tan \theta_3$ are the real roots of the equation $x^3 - (a+1)x^2 + (b-a)x - b = 0$, where $\theta_1 + \theta_2 + \theta_3 \in (0, \pi)$, then $\theta_1 + \theta_2 + \theta_3$ is equal to
- (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$
(C) $\frac{3\pi}{4}$ (D) π
12. Two circles with radii $\sqrt{2}$ cm and 1 cm meet at a point A, whose centres lie on x-axis. The distance between their centres is 2 cm. The chord AC of the larger circle cuts the smaller circle at a point B and is bisected by that point. The length of chord AC is not equal to
- (A) $\frac{\sqrt{5}}{2}$ cm (B) $\sqrt{\frac{7}{2}}$ cm
(C) $\frac{\sqrt{3}}{2}$ cm (D) $\sqrt{\frac{5}{2}}$ cm

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Let P is lying on x-axis and B(-2,4), C(-3,5) are given then the number of possible points P such that $PA + PB = 7$ is/are
2. In a triangle ABC, the least values of $\Pi \operatorname{cosec} \left(\frac{A}{2} \right)$ and $\sum \operatorname{cosec}^2 A$ are p and q respectively, then $(p - q)$ is equal to
3. A diagonal of rhombus ABCD is a member of both the family of lines $(x + y - 1) + \lambda_1(2x + 3y - 2) = 0$ and $(x - y + 2) + \lambda_2(2x - 3y + 5) = 0$ where $\lambda_1, \lambda_2 \in \mathbb{R}$ and one of the vertex of the rhombus is (3, 2). If area of the rhombus is $12\sqrt{5}$ square units then find the length of semi longer diagonal of the rhombus
4. Number of solution of the equation $\sin^4 x - \cos^2 x \sin x + 2 \sin^2 x + \sin x = 0$ in $0 \leq x \leq 3\pi$ is
5. The centres of the three circles A, B and C are collinear with the centre of circle B lying between the centres of circle A and C. Circles A and C both touch externally to circle B and the three circles share a common tangent line. Given that circle A has radius 12 and circle B has radius 42 the radius of circle C, is equal to
6. Curves $ax^2 + 2hxy + by^2 - 2gx - 2fy + c = 0$ and $a'x^2 - 2hxy + (a' + a - b)y^2 - 2g'x - 2f'y + c = 0$ intersect at four concyclic points A, B, C and D. If P is the point $\left(\frac{g' + g}{a' + a}, \frac{f' + f}{a' + a} \right)$, the value of $\frac{PA^2 + PB^2 + PC^2}{PD^2}$ is

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work

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E-24/24

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